

Disproof of Conjecture 00000003560

Brunn–Minkowski equality does not imply congruence

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Claim addressed. The source identifies the equality-case structure in Brunn–Minkowski with congruence of the summands. Without an additional equal-volume normalization, this is false even for positive-volume compact convex intervals. Congruence means an isometry of the ambient Euclidean space taking one summand onto the other.

Proposition 1. *Let $A = [0, 1]$ and $B = [0, 2]$ in $\mathbb{R}$. They are compact convex bodies with positive Lebesgue measure. They satisfy equality in the one-dimensional Brunn–Minkowski inequality, but they are not congruent.*

Proof. Their Minkowski sum is

$$A + B = \{a + b : a \in A, b \in B\} = [0, 3].$$

Indeed, every sum belongs to $[0, 3]$. Conversely, if $0 \leq z \leq 1$, write $z = z + 0$; if $1 < z \leq 3$, write $z = 1 + (z - 1)$. Each is an admissible decomposition.

Writing $|S|$ for Lebesgue measure, the Brunn–Minkowski inequality in dimension $d = 1$ is

$$|A + B|^{1/d} \geq |A|^{1/d} + |B|^{1/d}.$$

Here $|A| = 1$, $|B| = 2$ and $|A + B| = 3$, so equality holds exactly.

An isometry preserves distances and hence diameters. But

$$\text{diam } A = 1, \quad \text{diam } B = 2.$$

Therefore no Euclidean isometry maps A onto B . □

This example exhibits the essential distinction: the intervals are positively homothetic, since $B = 2A$, but they are not congruent. Their unequal sizes do not prevent equality in Brunn–Minkowski.

Scope. The contradiction concerns the source’s congruence conclusion for the original summands in exact equality cases. The source states no equal-volume hypothesis or convention of rescaling each summand first. The example does not refute a corrected result formulated up to translations and positive dilations, or a stability statement after separate volume normalization. It does not assess the additional assertion about a genus-dependent rate. Both intervals have nonempty interior and positive finite measure, so the example does not use zero-volume degeneracies.

Lean correspondence and reproduction

The formal sets `A` and `B` are actual real closed intervals. Minkowski addition is Mathlib’s pointwise addition of sets. The theorem `minkowski_sum` proves the set equality $A + B = [0, 3]$ by actual witnesses for membership in the sum.

The volumes use the genuine Lebesgue measure `volume`, evaluated through `Real.volume_Icc`. The definition `BMEquality` is precisely the displayed equality with exponent $1/1$ in dimension one. The proof also checks actual compactness, convexity and strict positivity of both volumes.

The definition `Congruent` requires a function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying Mathlib’s `Isometry` predicate and $f(A) = B$. The contradiction uses the actual metric diameter and the general theorem that an isometry preserves the diameter of the image. The additional theorem `are_homothetic` proves $B = 2A$ directly. Finally, `congruence_claim_false` negates the universal congruence conclusion for positive-volume compact convex sets satisfying the equality.

Inside `lean/`, run `lake build`, followed by

```
lake env lean -DwarningAsError=true Main.lean.
```

Lean is pinned to 4.19.0 and Mathlib to

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

The manifest pins all transitive dependencies. No auxiliary numerical computation is needed. Actual build logs, axiom dependencies, hashes, and PDF checks are included in `verification/`. Author self-review and delegated-agent adversarial review are recorded separately; no external independent review is claimed.

References

- [1] The Last Math Competition, Conjecture 00000003560 (original bilingual source).